

PILOT v2 — “Variance components speak biology”

Review artifact, not shipped. Lives in docs/dev-log/pilots/ (never docs/articles/); changes nothing in the package. Numbers from real lmer / glmer fits. This v2 folds in your review: **both visual options** shown for you to pick, quantity labelled **ICC**, and the **binomial latent-scale ICC** added ($\pi^2/3$ logit, 1 probit). General latent-scale ICC (Poisson etc.) stays for later.

A. Gaussian random intercept — repeatability of boldness

boldness scored 4× on each of 30 lizards: boldness ~ assay_temp + (1 | lizard).

component	variance	SD	% of total
between lizards	1.15	1.07	63.9%
within (residual)	0.65	0.806	36.1%

Where does the variation live? — two visual options, pick one

Option A — single stacked bar (compact; ideal for between/within):

between lizards 63.9%

within (residual) 36.1%

Option B — one bar per component (a small bar chart; scales to 3+ components, e.g. site + plot + residual in nested designs):

between lizards

within (residual)

63.9% ($\sigma^2=1.15$)

36.1% ($\sigma^2=0.65$)

Of the total variation in boldness, about 63.9% sits between lizards and about 36.1% within them.

ICC line

ICC = 0.639. Lizards account for 63.9% of the variation; two boldness scores from the same lizard are correlated about this strongly. (In a single-random-intercept Gaussian design the ICC is the repeatability.)

$$\text{ICC} = \frac{\sigma_{\text{lizard}}^2}{\sigma_{\text{lizard}}^2 + \sigma_{\varepsilon}^2} = \frac{1.15}{1.8} = 0.639$$
 — a true proportion of variance because the Gaussian residual is on the same (data) scale.

Each $\hat{u}(\text{lizard})$ is partially pooled — nudged toward the overall mean; lizards with fewer/noisier scores pulled harder. That borrowing of strength is what $(1|\text{lizard})$ does. (prose only; numeric shrinkage deferred)

B. Binomial random intercept — latent-scale ICC (NEW: feasible now)

nestling survival (0/1), 6 chicks in each of 40 nests: $\text{surv} \sim \text{temp} + (1 | \text{nest}), \text{logit link}.$

component	variance (logit scale)	SD
between nests	1.25	1.12
latent residual (fixed: $\pi^2/3$)	3.29	1.81

Latent-scale ICC (logit link)

between nests 27.5%

latent residual ($\pi^2/3$) 72.5%

Latent-scale ICC = 0.275 (logit scale). On the unobserved logit scale, nests account for 27.5% of the variation. The residual variance on this scale is the *known* logistic constant $\pi^2/3 \approx 3.29$ (it would be 1 for a probit link $\rightarrow \text{ICC} = 0.556$). **This is NOT a proportion of variance in the observed 0/1 survival** — a Bernoulli response has no single data-scale residual variance, so only the latent-scale reading is honest here.

$$\text{ICC}_{\text{latent}} = \frac{\sigma_{\text{nest}}^2}{\sigma_{\text{nest}}^2 + \pi^2/3} = \frac{1.25}{4.54} = 0.275$$

C. How it integrates with the existing functions

explain(fit) — current output (excerpt):

- The new “How the variation splits” panel (chosen bar + ICC line) slots in HERE, after the random effects, reusing the existing knit_print for the table.
- `model_card(fit)` gains the same panel.
- Three-views widget, **Index tab**: the chosen bar + one sentence beneath the random-effects glossary (plain CSS divs — no JS, survives PDF).
- `variance_partition(sym)` & `icc(sym)`: new accessors beside `group_means()` / `group_slopes()` (same family-gate + honest-print contract).

D. Honesty contract

- **Gaussian-identity** single random intercept → data-scale ICC & % (a true variance proportion). ✓
- **Binomial (logit/probit)** → **latent-scale** ICC with the fixed residual ($\pi^2/3$ or 1), always explicitly labelled “latent scale, not a proportion of observed-outcome variance.” ✓ (this v2)
- **Poisson / other GLMMs** → deferred (need the log-normal / trigamma distribution-specific variance). Table shows; ICC line withheld with a one-line “not yet on this scale” note. —
- Every reading: point-estimate-only caveat + reuse `few_re_levels` / `few_groups_wald` warnings; all prose from `variance-readings.csv`.